

MUHAMMED AMAN SHAMINAS

Kannur, Kerala, India | 8891284295 | Amanshamnas2@gmail.com
linkedin.com/in/muhammed-aman-809973383 | github.com/muhammed-aman77

PROFESSIONAL SUMMARY

Final-year B.E. student in Artificial Intelligence & Machine Learning (CGPA 7.16/10) with hands-on project experience in embedded sensing, machine learning, and web development (Python, React, Next.js). Developed an ESP32-based TinyML machine health monitoring system and the front-end of a team restaurant application. Seeking an AI/ML, software development, or front-end internship or entry-level role.

TECHNICAL SKILLS

Programming Languages: Python, C, JavaScript

Web Development: HTML, CSS, React, Next.js, Node.js, REST APIs, Streamlit

AI / Machine Learning: Machine Learning, Image Processing, Feature Engineering, scikit-learn, TinyML, TensorFlow Lite

Embedded / IoT: ESP32, Arduino, I2C and Serial Communication, Sensor Interfacing (accelerometer, temperature, sound, current)

Databases: MySQL, MongoDB

Tools: Git, GitHub, VS Code, Figma, Excel, Tableau

PROJECTS

TinyML-Based Machine Health Monitoring System

2025 – Present

- Collected sensor data (vibration, temperature, sound) to monitor machine condition in real time.
- Deployed a lightweight ML model on an ESP32 for low-power edge inference without cloud processing.
- Enabled early fault detection to support predictive maintenance.

Daily Dine – Restaurant Application

2025 – Present

- Front-End Developer in a team of 7–8; designed responsive UI screens in Figma and implemented them in Next.js.
- Developed customer-facing digital menu interfaces to make browsing and ordering easier for restaurant customers.

Smart Fire Detection & Battery Safety System for EVs

2026

- Reviewed 10 research papers and patents on lithium-ion thermal runaway, battery management, and EV fire safety to identify gaps in existing solutions.
- Prototyping on Arduino UNO with DS18B20 temperature sensing, ACS712 current sensing, relay-based power cutoff, LCD display, buzzer/LED alerts, and SIM800L GSM alerts to the vehicle owner.

AI-Based Diabetic Detection Using Eye Images

2025

- Developed a machine learning mini project for early diabetes detection from retinal images.
- Applied image preprocessing and feature extraction, then evaluated model performance on held-out data.

EDUCATION

B.E. in Artificial Intelligence & Machine Learning, VTU

2023 – 2027

Srinivas Institute of Technology, Mangaluru | CGPA: 7.16/10

Higher Secondary, Kerala State Board

2023

Kadambur Higher Secondary School, Kannur, Kerala | 93.5%

SSLC, Kerala State Board

2021

Kadambur Higher Secondary School, Kannur, Kerala | 100%

CERTIFICATIONS

- AWS Academy Graduate – Cloud Architecting, Amazon Web Services (2026, 60 hrs)
- AI Tools & ChatGPT Workshop, be10X (Sep 2026)
- Market Research: Research Methodology, Infosys (2025)

ACHIEVEMENTS & ACTIVITIES

- 2nd place, college-level technical event (Error Fixing) – debugging and problem-solving.
- 1st place, inter-college football tournament – teamwork and leadership.
- Volunteer and event organizer for the college tech fest Altius; designed posters and promotional materials.

Languages: English, Hindi, Malayalam